2019-20
Half-yearly Exam
S.1 MATHS

PAPER 2

QUEEN'S COLLEGE

HALF-YEARLY EXAMINATION 2019 – 2020
SECONDARY 1 MATHEMATICS
PAPER 2
Question Book

Date:	17^{th} January 2020
Time:	08:30 a.m. – 09:30 a.m.
Time Allowed:	60 minutes
Max. Mark:	80

INSTRUCTIONS

1. Read carefully the instructions on the Answer Sheet. Insert the information required in the spaces provided on Answer Sheet.

2. When told to open this book, you should check that all questions are there. Look for the words '**END OF PAPER**' after the last question.

3. All questions carry equal marks.

4. **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly.

5. The diagrams in this paper are not necessarily drawn to scale.

6. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

7. No marks will be deducted for wrong answers.

8. Calculators are **NOT** allowed in this paper.

There are 40 questions in this paper.
All questions carry equal marks.
Choose the best answer for each question.

1. Find the sum of the first 3 consecutive prime numbers.
   A. 6
   B. 10
   C. 12
   D. 30

2. Express 504 in index notation form.
   A. $2^3 \times 3^2 \times 7$
   B. $2^2 \times 3^3 \times 7$
   C. $2^2 \times 3^2 \times 7$
   D. $2 \times 3^3 \times 7$

3. Find the highest common factor of $2^2 \times 3^3 \times 7$, $2^3 \times 3^2 \times 5$ and $2 \times 3^2 \times 5 \times 7$.
   A. 12
   B. 18
   C. 1260
   D. 2520

4. How many negative integers between -3.7 and 3.2?
   A. 3
   B. 4
   C. 5
   D. 6

5. Which of the following statements are correct?
   I. Zero is a natural number.
   II. Zero is an integer.
   III. Zero is a positive number.
   IV. Zero is an even number.

   A. I and III only
   B. II and IV only
   C. II, III and IV only
   D. I, II and IV only

6. Arrange the following numbers in descending order.

$-40.7,\ -3\frac{2}{3},\ -16.3,\ -\frac{120}{3}$

A. $-3\frac{2}{3} > -16.3 > -40.7 > -\frac{120}{3}$

B. $-3\frac{2}{3} > -16.3 > -\frac{120}{3} > -40.7$

C. $-40.7 > -\frac{120}{3} > -16.3 > -3\frac{2}{3}$

D. $-\frac{120}{3} > -40.7 > -16.3 > -3\frac{2}{3}$

7. $(+3)(-6) \div (-1) =$

A. -18

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. 18

8. $\left(-3\frac{1}{7}\right)\left(+\frac{2}{11}\right) + \left(-6\frac{1}{4}\right) \div \left(-3\frac{1}{2}\right) =$

A. $-\frac{33}{14}$.

B. $-\frac{17}{14}$.

C. $\frac{17}{14}$.

D. $\frac{33}{14}$.

9. Which of the following statements about the number line must be true?

I. The number zero must be included

II. The directed number placed horizontally from right to left in descending order.

III. The directed number placed vertically from top to bottom in descending order.

IV. The directed number placed vertically from bottom to top in descending order.

A. I and II only

B. II and III only

C. I, II and III only

D. I, II and IV only

10. In a certain country, the temperature in a winter morning is $-6\,°\text{C}$ and then it drops to $-23\,°\text{C}$ at night. Find the temperature change from the morning to the night.

A. $-29\,°\text{C}$

B. $-17\,°\text{C}$

C. $17\,°\text{C}$

D. $29\,°\text{C}$

11. Evaluate $[-2^2]^2 \div 4^2 \times [-2^2]^2 \div 4^2$.

A. 2

B. 1

C. -1

D. -2

12. Simplify $2(3-4\times p \div 2 \times 6q)$.

A. $6-\dfrac{p}{3q}$

B. $6-\dfrac{p}{6q}$

C. $6-48pq$

D. $6-24pq$

13. Find the constant term after simplifying the expression $5x\times 4+(-3)\times 2+7(-4+x)$.

A. 34

B. 22

C. -22

D. -34

14. Solve $\dfrac{5x-3}{6}=\dfrac{x}{3}-5$.

A. $x=-11$

B. $x=-9$

C. $x=-\dfrac{7}{3}$

D. $x=-\dfrac{2}{3}$

15. Solve $\frac{2}{3}(x+5)-\frac{1}{6}=-\frac{5x}{2}$.

A. $x=1$

B. $x=\frac{4}{19}$

C. $x=-\frac{4}{19}$

D. $x=-1$

16. Represent the statement "Subtract the product of the cube of r and q from 8" by an algebraic expression.

A. $8-r^3q$

B. $8-r^2q$

C. r^3q-8

D. r^2q-8

17. In a restaurant, the cost of a set dinner is $132 per person. If the owner wants to make a profit of $1620 from a table of 6. How much is a set dinner for each person?

A. $402

B. $292

C. $270

D. $248

18. The price of an adult ticket is $ x and the price of a child ticket is one third of an adult ticket. If $510 are paid for 7 adults and 13 children. How much is an adult ticket?

A. $15

B. $25.5

C. $45

D. $76.5

19. Ben and Jacky own no more than 50 comics books altogether. If Ben owns y comics books and Jacky owns 1.5 times that of Ben, which of the following is NOT a possible value of y ?

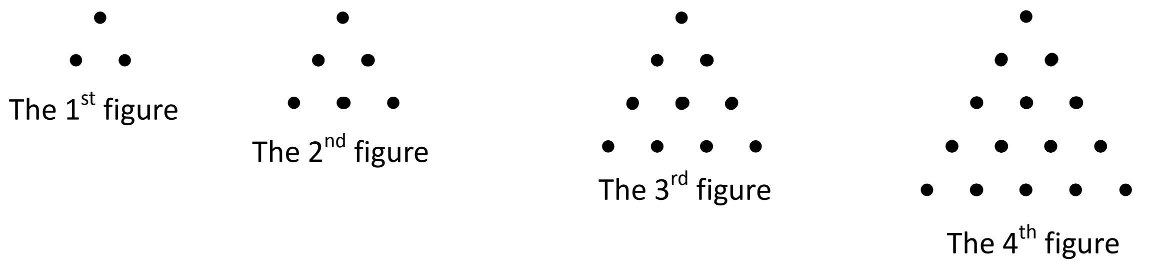
A. 18

B. 19

C. 20

D. 21

20. The below figures are formed by identical matches.



What kind of sequence can be formed by the number of matches in each figure?

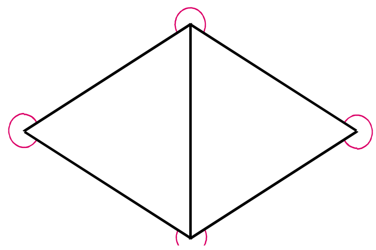
A. Sequence of Triangular numbers
B. Arithmetic sequence
C. Geometric sequence
D. Fibonacci sequence

21. Consider the sequence $2, 8, 18, 32 \ldots\ldots$
If the k th term is 162, find the value of k.
A. 7
B. 8
C. 9
D. 10

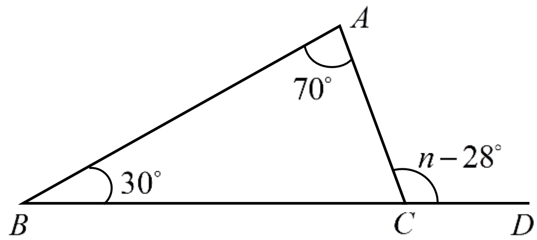
22. Which of the following is / are correct?
I. An angle larger than 0° and smaller than 90° is an acute angle.
II. An angle larger than 90° and smaller than 270° is an obtuse angle.
III. An angle larger than 270° and smaller than 360° is a reflex angle.

A. I only
B. I and II only
C. I and III only
D. I, II and III

23. In the figure, find the sum of all marked angles.
A. 720°
B. 900°
C. 1080°
D. 1260°



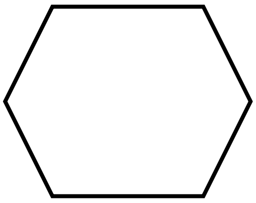
24. In the figure, BCD is a straight line and $\angle ACD = n - 28^\circ$. Find n.
A. 72°
B. 98°
C. 128°



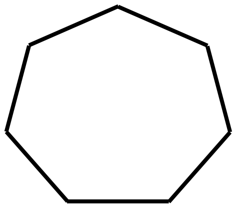
D. 152°

25. Which of the following pairs of polygon and name do not match?

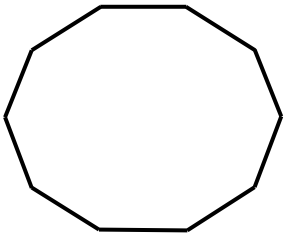
A. Hexagon



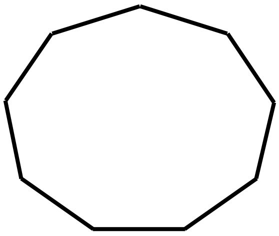
B. Heptagon



C. Octagon

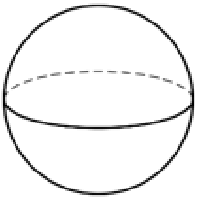


D. Nonagon



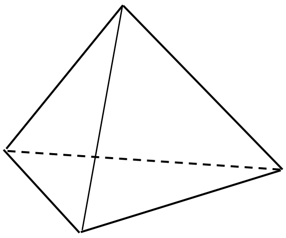
26. Which of the following is a hexahedron?

A.

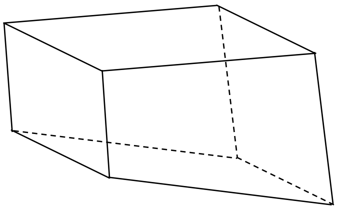


B.

C.



D.



27. Sammy is 126 cm tall. If her height is 75% of that of her elder brother's, Sammy's brother is taller than her by

A. 12 cm.

B. 22 cm.

C. 32 cm.

D. 42 cm.

28. Which of the following is/ are correct?

I. $200\% + 5\% = 205$

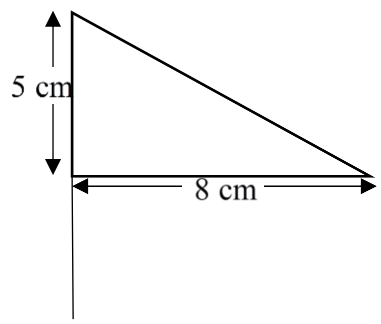
II. $200\% - 50\% = 1.5$

III. $200\% \times 50\% = 10$

IV. $200\% \div 5\% = 40$

A. I and IV only
B. II and III only
C. II and IV only
D. III and IV only

29. In the figure, the base and the height of the triangular flag are 8 cm and 5 cm respectively. If the height and the base increased by 20% and 25% respectively, what is the percentage increase in the area of the flag?



A. 33.3%
B. 50%
C. 133.3%
D. 500%

30. The kitchen kit is marked 20% above the cost price and sold at a 25% discount in the boxing day. Princess bought the kitchen kit for $900, what is the cost price of the kit?
A. $1440
B. $1406
C. $1275
D. $1000

31. Kenny bought 280 glasses online for $4905, 15% of them are broken during the transportation. If he sells the rest of them at $45 for a set of 2 glasses. His overall
A. gain is $5805.
B. gain is $450.
C. loss is $3015.
D. loss is $3960.

32. There are 2 paths on the mountain. The distance of path A and path B are 206.54 km and 2356.98 m respectively. Estimate the number of times of the distance of path B is that of path A.
A. 0.01
B. 100
C. 10
D. 10000

33. The table below listed the monthly expenditure of Sandy.

Expenditure	Fee
Rent	$ 8670
Electricity	$ 1390

Water	$ 730
Groceries	$ 520
Transportation	$ 245

Estimate the minimum monthly salary of Sandy in order to pay the expenditure?

A. $11800

B. $11500

C. $11400

D. $11100

34. Wong's family spent $255, $312, $465, $111 and $596 in last weekend. If every $1 spend earns 1 point and every 50 points may redeem 1 coupon. Mr Wong would like to calculate the maximum number of coupons that he can redeem according to the expenditure in last weekend. Which of the following estimation must be correct?

Method	Maximum number of coupons
A. Front-end	35
B. Round-off each number to the nearest ten	34
C. Round-down each number to the nearest ten	34
D. Round-up each number to the nearest ten	35

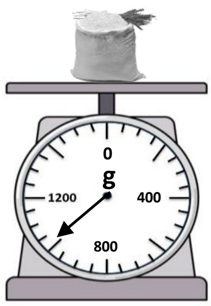
35. In the figure, it is the scale reading of a pack of flour.
What is the weight of the flour?

A. 1100 g (corr. to the nearest 100 g)

B. 1050 g (corr. to the nearest 100 g)

C. 1050 g (corr. to the nearest 50 g)

D. 1000 g (corr. to the nearest 50 g)



36. Estimate the value of $351.8-28.8\div2.4\times12.4$ by front-end method.

A. 206

B. 203

C. 200

D. 182

37. The L.C.M. and H.C.F. of three numbers *A*, *B* and *C* are $2^{31}\times3^{6}\times5^{4}\times13$ and $2^{17}\times5^{2}\times13$ respectively. If $A=2^{31}\times3\times5^{3}\times13$ and $B=2^{17}\times3^{6}\times5^{2}\times13$, then $C=$

A. $2^{17}\times5^{4}\times13$.

B. $2^{17}\times3^{6}\times5^{4}\times13$.

C. $2^{31}\times5\times13$.

D. $2^{31}\times3\times5^{4}\times13$.

38. Karl has spent not less than $84 for snacks. He bought 6 boxes of chocolate and 5 packs of candies. If the price of a box of chocolate is $\$x$, and which is $3 more than the price of a pack of candy. What is the least possible integral price for a box of chocolate?

A. $8

B. $9

C. $10

D. $11

39. The general term of the sequence $\frac{9}{4},-\frac{27}{8},\frac{81}{16},-\frac{243}{32}$ is

A. $(-\frac{3}{2})^{n+1}$.

B. $-(\frac{3}{2})^{n+1}$.

C. $-\frac{3n+1}{2^{n}}$.

D. $\frac{3n+1}{2^{n}}$.

40. The marked price of a printer is $y\%$ higher than its cost. During Thanksgiving Day, there is 25% off to the printer and is sold at $1260. The loss percentage in selling a printer is 10%. Find the value of y.

A. 12

B. 13.6

C. 20

D. 21.2

END OF PAPER